



UNIVERSITY
OF COLOGNE

BRACKETS, TREES AND THE BORROMEAN RINGS

Gustavo Jasso

A bird's-eye view on algebraic structures

Mathematical objects



Algebraic structures

X : topological space

X : smooth manifold

V : complex variety

G : Lie group

$\pi_k(X, x)$: homotopy groups

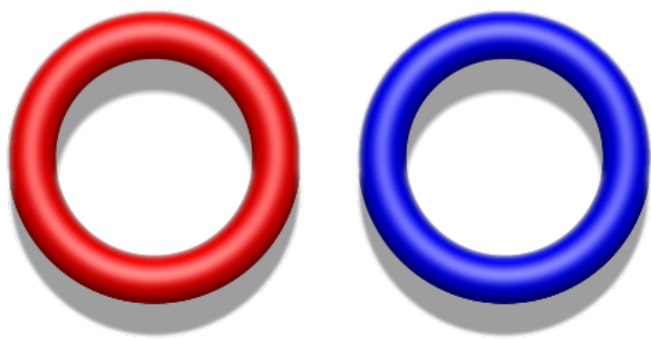
$\Omega(X)$: algebra of differential forms

$\mathbb{C}[V]$: coordinate ring

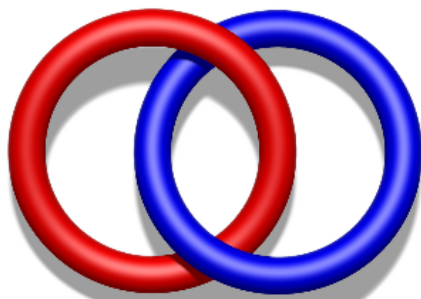
$\text{Lie}(G)$: Lie algebra

Properties of the mathematical objects
should be reflected in properties of the algebraic structures

What differentiates these configurations of each other?

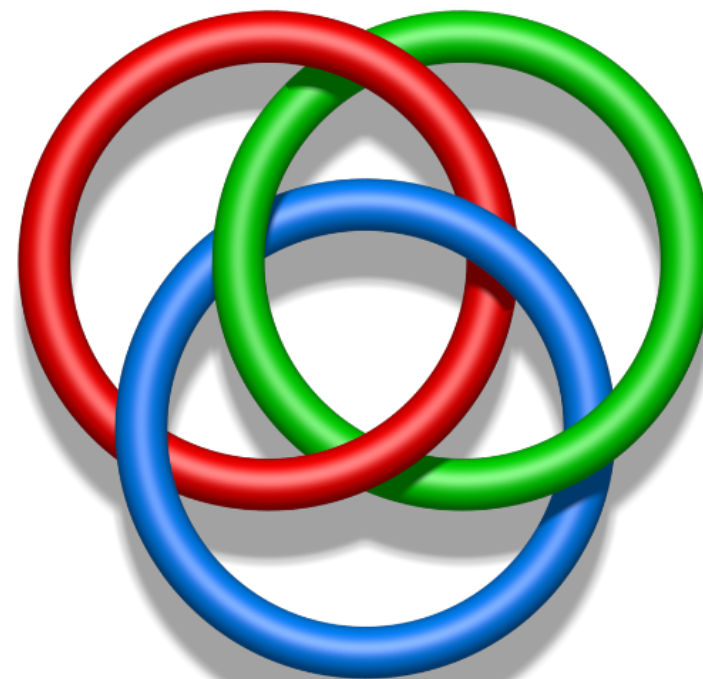


Hopf link



The Hopf link is, well, linked!

Consider now the Borromean rings:



What differentiates them from three unlinked circles?

01

BRACKETS AND TREES



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The combinatorics of the associativity equation

The familiar associativity equation

$$(ab)c = a(bc)$$

is the source of rich combinatorics:

1 : (ab)

2 : $((ab)c), (a(bc))$

5 : $((ab)c)d), ((a(bc))d), (a((bc)d))$
 $(a(b(cd))), ((ab)(cd))$

14 : $((((ab)c)d)e), (((a(bc))d)e), \dots$

Thm (Catalan 1838)

There are precisely

$$C_n := \frac{1}{n+1} \binom{2n}{n}$$

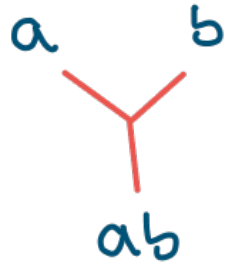
different ways to correctly
parenthesise a word on
 $n+1$ letters.

$$C_0 := 1, \quad C_{n+1} := \sum_{k=0}^n C_k C_{n-k}$$

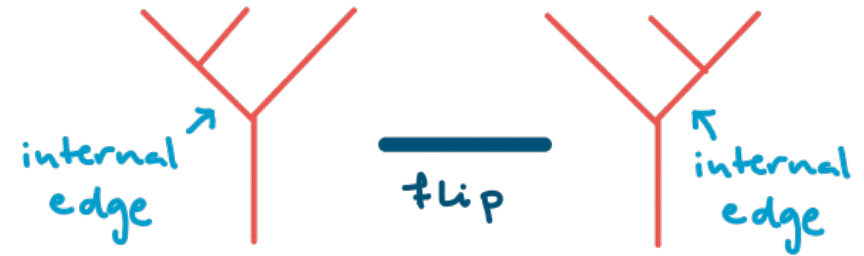


From associativity to dendrology

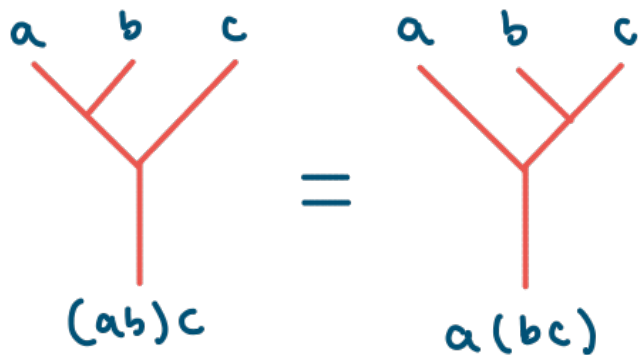
Depict the binary operation as a tree



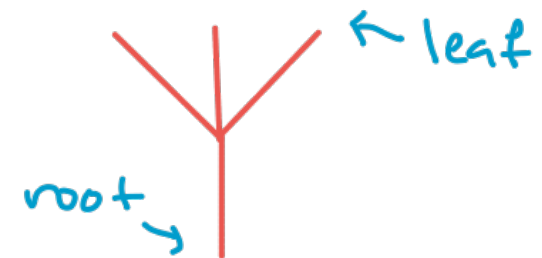
This suggests the local move:



The associativity eq'n becomes



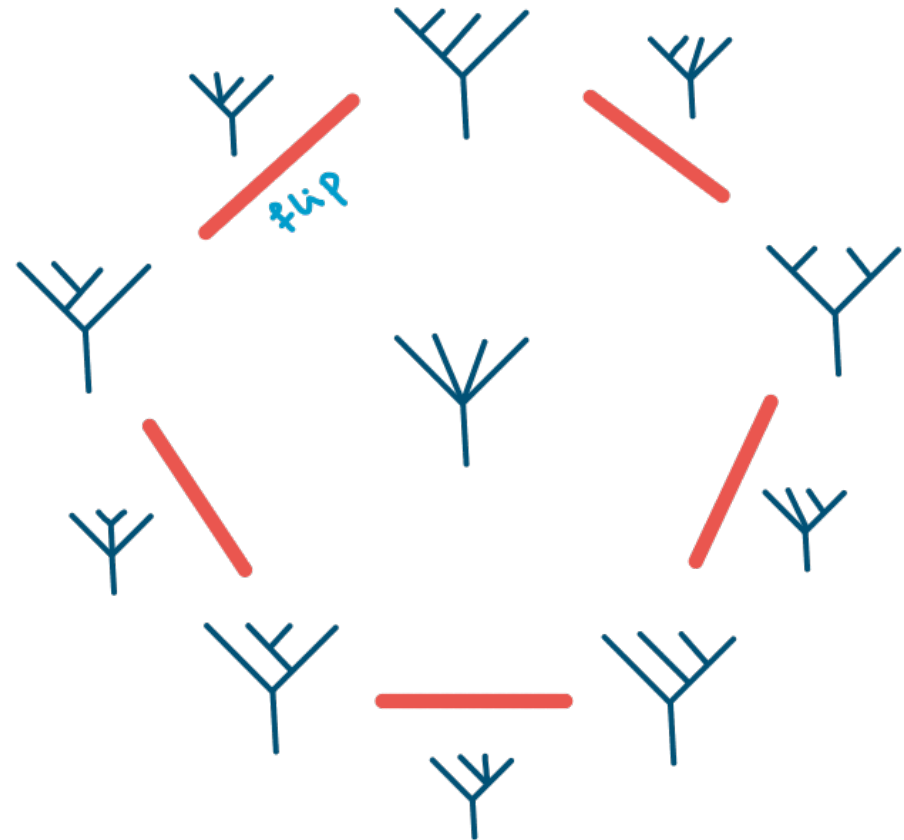
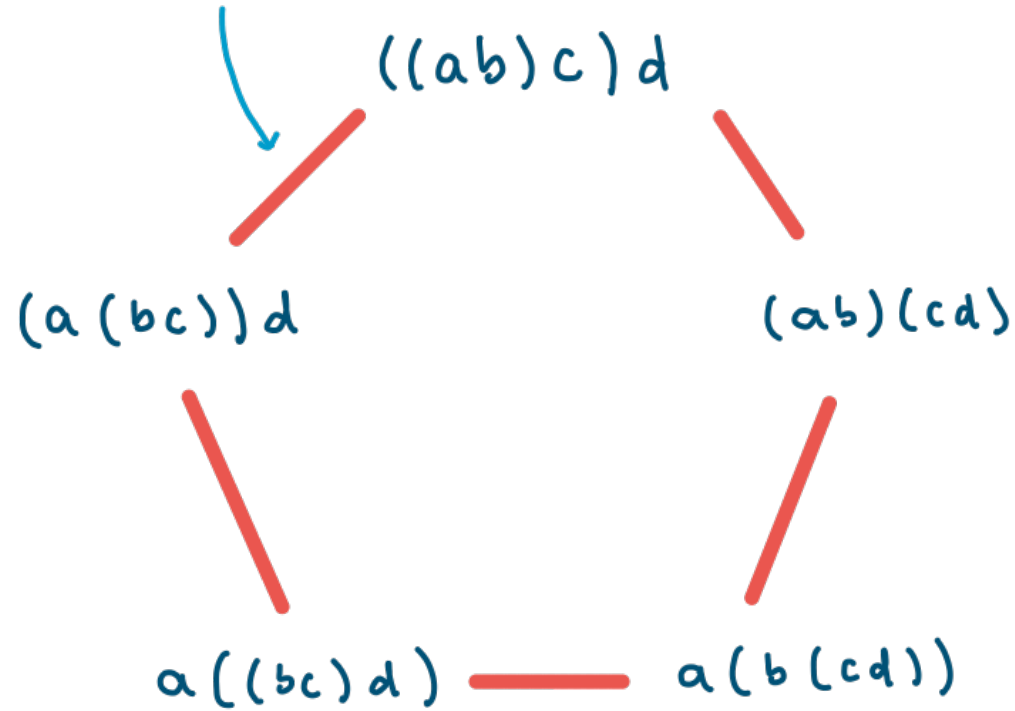
Contracting the internal edges:



The missing planar tree
with three leaves

From associativity to dendrology II

associativity law



The Associahedron: "A mythical polytope"

Thm (Tamari 1951, Stasheff 1963, Loday 2004)

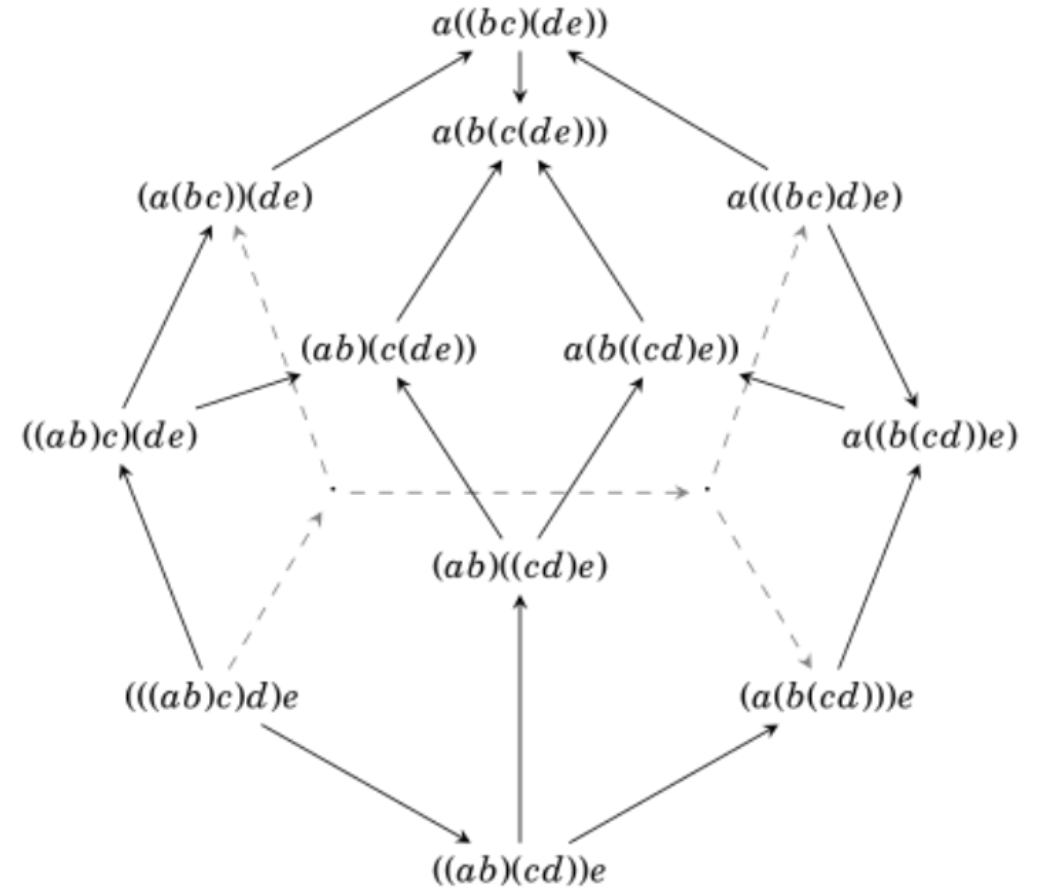
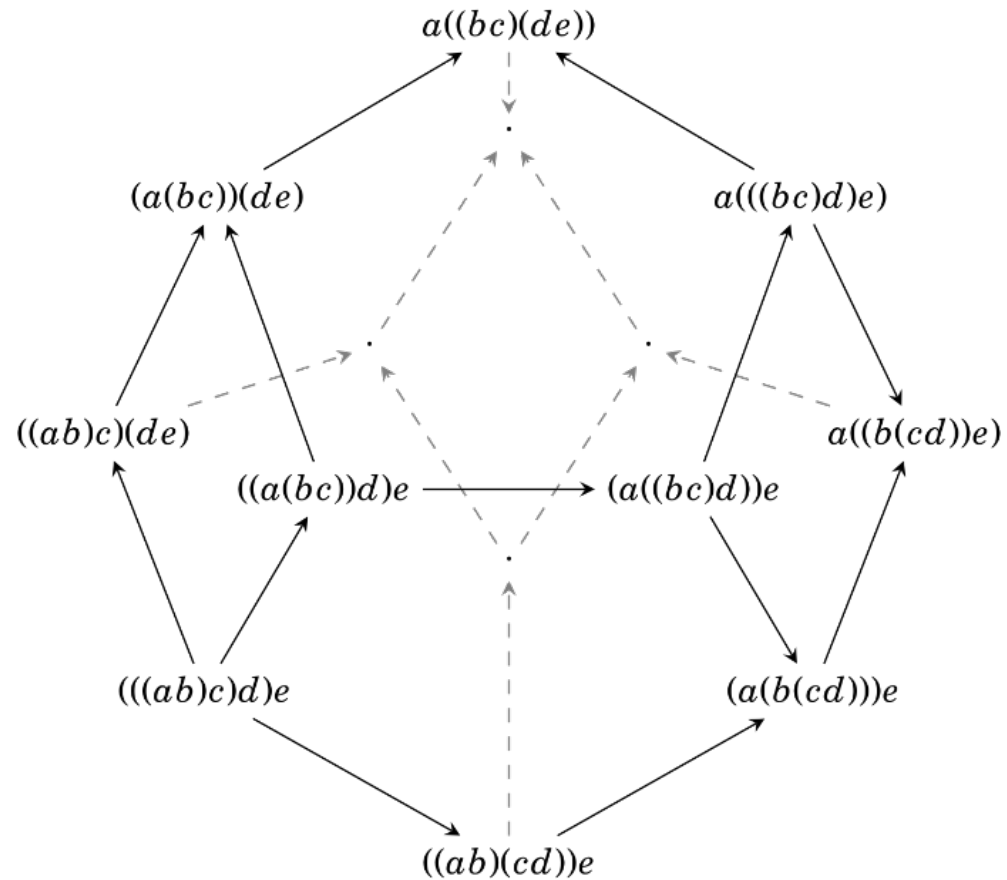
$\exists$ convex polytope K_{n+1} of dim $n-1$ st.

$$\left\{ \begin{array}{l} k\text{-dim cells} \\ \text{of } K_{n+1} \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{l} \text{planar rooted trees with} \\ n+1 \text{ leaves and } n-k \text{ int. vertices} \end{array} \right\}$$

called the Associahedron of dim $n-1$

Rmk Edges of the associahedron correspond to flips!

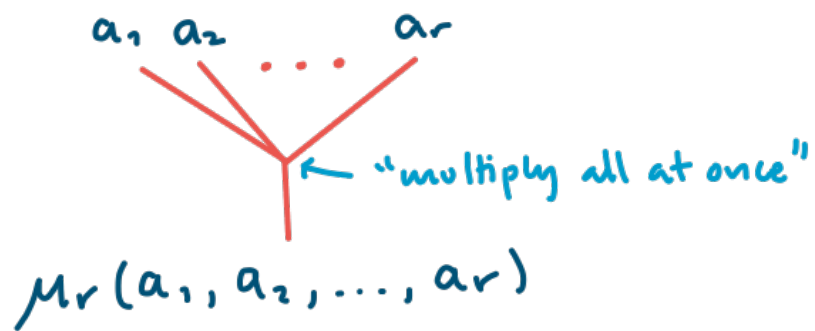
The Associahedron K_5



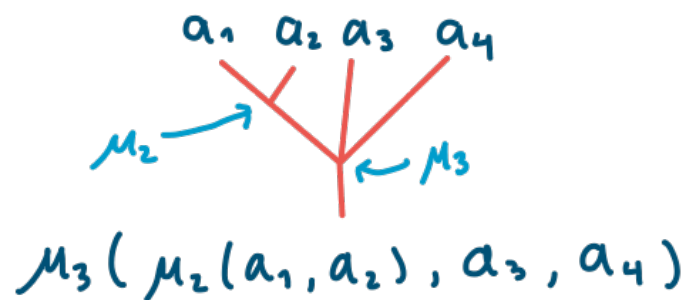
Planar trees as multi-operations

Rooted corolla with r leaves

$\rightsquigarrow$ operation with r inputs



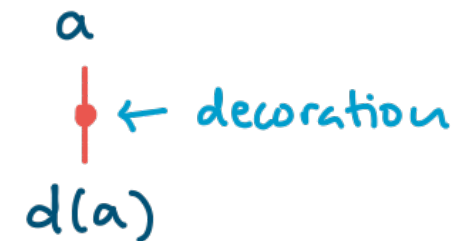
Combinations of these yield
new multi-operations



We also consider a unary operation

$$a \mapsto \mu_1(a) =: d(a)$$

corresponding to the tree



Remark The $\bullet$ indicates that something happens as a flows through the tree

The A_∞ -equations (Stasheff 1963)

$$\begin{array}{c} a \\ | \\ \bullet \\ | \\ \bullet \\ | \\ d^2(a) \end{array} = 0$$

$$\begin{array}{c} a \quad b \\ \diagdown \quad \diagup \\ \bullet \end{array} - \begin{array}{c} a \quad b \\ \diagdown \quad \diagup \\ \bullet \end{array} - \begin{array}{c} a \quad b \\ \diagdown \quad \diagup \\ \bullet \end{array} = 0$$

$$d(ab) \quad d(a)b \quad ad(b)$$

Leibniz Rule

$$\begin{array}{c} a \quad b \quad c \\ \diagdown \quad | \quad \diagup \\ \bullet \end{array} + \begin{array}{c} a \quad b \quad c \\ \diagdown \quad \bullet \quad \diagup \\ | \end{array} + \begin{array}{c} a \quad b \quad c \\ \diagdown \quad | \quad \bullet \end{array} + \begin{array}{c} a \quad b \quad c \\ \diagdown \quad \bullet \quad | \end{array} = \begin{array}{c} a \quad b \quad c \\ \diagdown \quad \diagup \\ | \end{array} - \begin{array}{c} a \quad b \quad c \\ \diagup \quad \diagdown \\ | \end{array}$$

$$\underbrace{d(\mu_3(a,b,c)) \quad \mu_3(d(a),b,c) \quad \mu_3(a,d(b),c) \quad \mu_3(a,b,d(c))}_{\partial(\mu_3)(a,b,c)} = \underbrace{a(bc) \quad (ab)c}_{\text{associator of } \mu_2}$$

$$\partial\left(\begin{array}{c} \diagup \quad \diagdown \quad \diagup \\ | \end{array}\right) = - \begin{array}{c} \diagup \quad \diagdown \quad \diagup \\ | \end{array} + \begin{array}{c} \diagup \quad \diagdown \quad \diagup \\ | \end{array} - \begin{array}{c} \diagup \quad \diagdown \quad \diagup \\ | \end{array} + \begin{array}{c} \diagup \quad \diagdown \quad \diagup \\ | \end{array} + \begin{array}{c} \diagup \quad \diagdown \quad \diagup \\ | \end{array}$$

etc.

A simple example of an A_∞ -structure

$\varphi: \mathbb{C}^n \rightarrow \mathbb{C}^n$ linear map s.t. $\varphi^{\ell} = 0$ ← $\ell \geq 3$ fixed

$(\mathbb{C}^n, \varphi)$: representation of $\mathbb{C}[x]/(x^{\ell}) =: A$

$$\text{JNF } \varphi \cong \text{J}_0(n_1) \oplus \dots \oplus \text{J}_0(n_k)$$

$$n = n_1 + \dots + n_k$$

Simple(st) representation: $S = (\mathbb{C}, \boxed{\circ})$

$$\text{J}_0(n) = \begin{pmatrix} \boxed{\circ} & 1 & & 0 \\ & \boxed{\circ} & \ddots & \\ 0 & & \ddots & 1 \\ & & & \boxed{\circ} \end{pmatrix} \text{ is "built" from } S$$

The "building procedure" is encoded in the Yoneda algebra:

$$\text{Ext}_A^*(S, S) \cong \mathbb{C}[\varepsilon, t]/(\varepsilon^2) \left. \vphantom{\begin{matrix} \text{Ext}_A^*(S, S) \cong \mathbb{C}[\varepsilon, t]/(\varepsilon^2) \\ |\varepsilon|=1 \quad |t|=2 \end{matrix}} \right\} \begin{matrix} \text{independent} \\ \text{of } \ell \end{matrix}$$

$\mu_1 = 0$ $\mu_2 = \text{Yoneda product}$

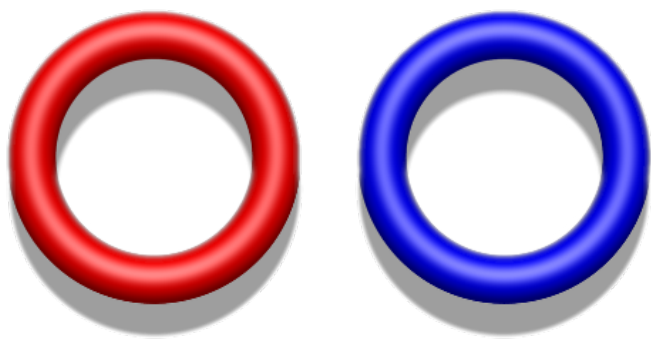
$\mu_{\ell}(\varepsilon, \dots, \varepsilon) = t$

$\mu_k = 0$ for $k \neq 2, \ell$

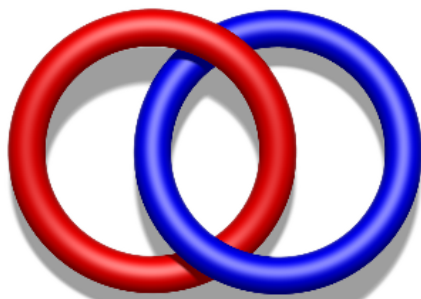
02

BACK TO THE BORROMEAN RINGS

What differentiates these configurations of each other?

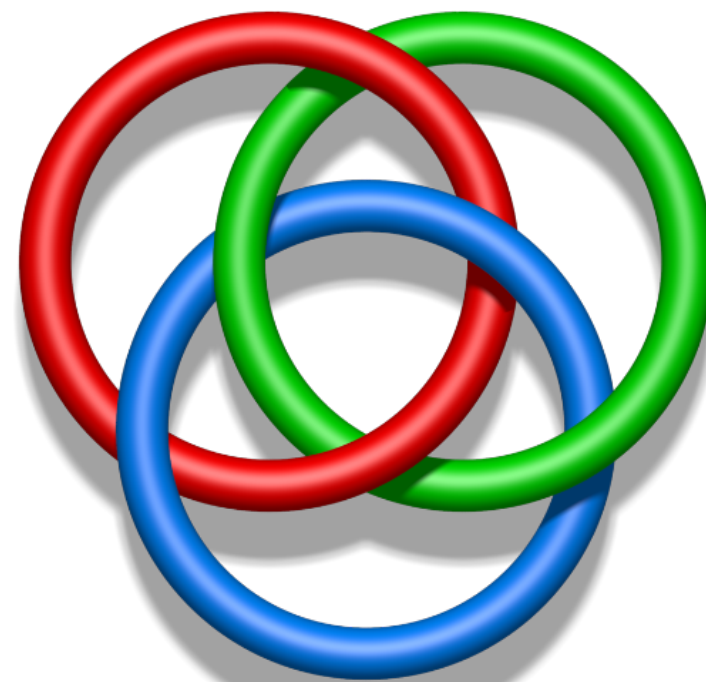


Hopf link



The Hopf link is, well, linked!

Consider now the Borromean rings



What differentiates them from three unlinked circles?



Move on the Hopf link

$$X := S^3 \setminus \text{Hopf link}$$

The singular cohomology of X is

$$H^0(X; \mathbb{R}) = \mathbb{R}$$

$$H^1(X; \mathbb{R}) = \mathbb{R}\alpha \oplus \mathbb{R}\beta$$

$$H^2(X; \mathbb{R}) = \mathbb{R}$$

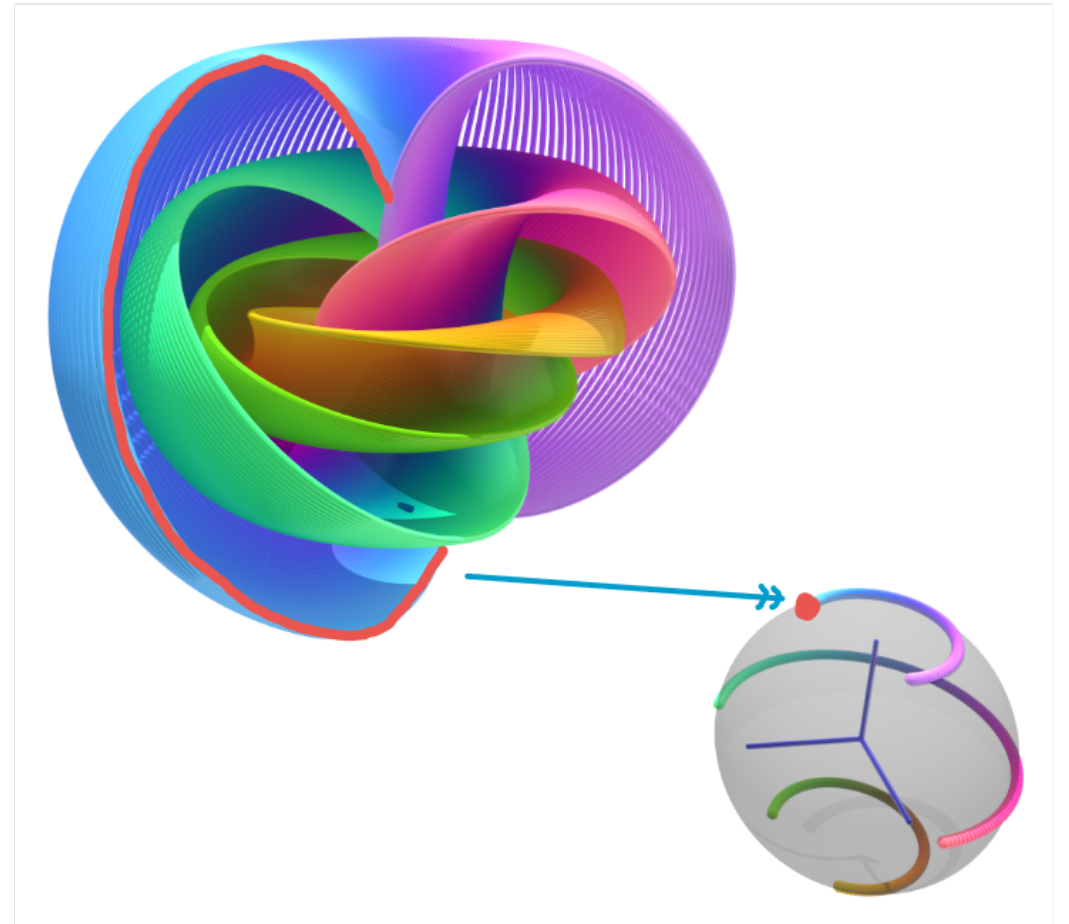
with the cup product $\mu_2^X(\alpha, \beta) \neq 0$

$$Y := S^3 \setminus \text{two unlinked circles}$$

identical cohomology but $\mu_2^Y(\alpha, \beta) = 0$

The Hopf fibration

$$\begin{aligned} \mathbb{C}^2 \cong S^3 &\longrightarrow S^2 \subseteq \mathbb{C} \times \mathbb{R} \\ (w, z) &\mapsto (2w\bar{z}, |w|^2 - |z|^2) \end{aligned}$$



A_∞ -structure on singular cohomology

X : top. space

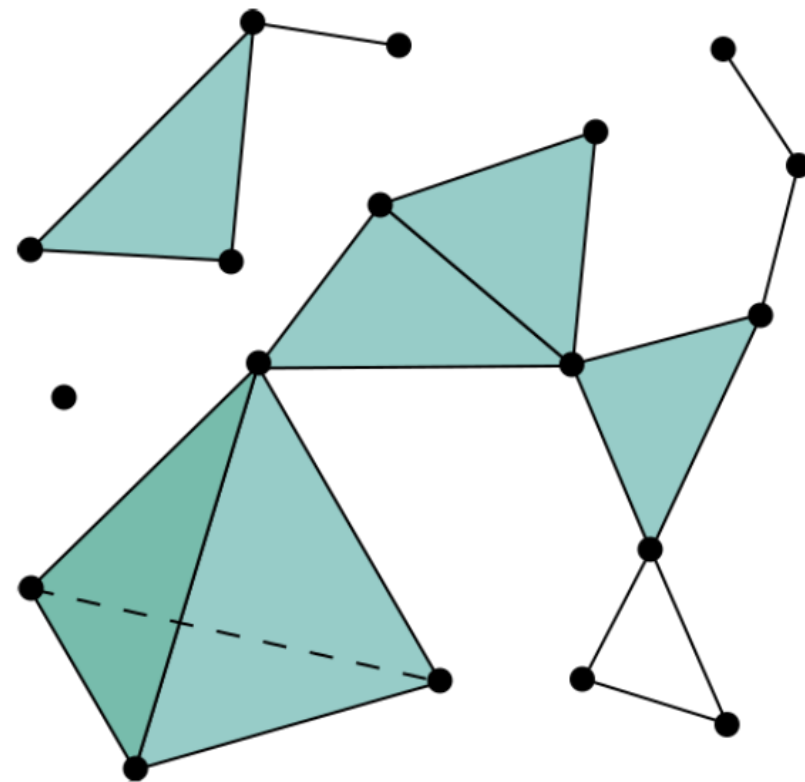
$$H^*(X; \mathbb{R}) := \bigoplus_{p=0}^{\infty} H^p(X; \mathbb{R})$$

admits operations

$$d := \mu_1, \mu_2, \mu_3, \dots$$

that satisfy the A_∞ -equations, where

$d = 0$ and μ_2 : cup product



What do the higher operations tell us about the space X ?

Massey's beautiful theorem

$$X := S^3 \setminus \text{Borromean rings}$$

$$H^0(X; \mathbb{R}) = \mathbb{R}$$

$$H^1(X; \mathbb{R}) = \mathbb{R}\alpha \oplus \mathbb{R}\beta \oplus \mathbb{R}\gamma$$

$$H^2(X; \mathbb{R}) = \mathbb{R}$$

$$\mu_2^x(\alpha, \beta) = \mu_2^x(\alpha, \gamma) = \mu_2^x(\beta, \gamma) = 0$$

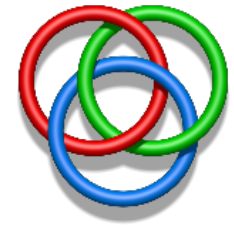
The same is true for

$$Y := S^3 \setminus \text{three unlinked circles}$$

Thm (Massey 1969)

$$\text{For } X := S^3 \setminus \text{Borromean rings}$$

$$\mu_3^x(\alpha, \beta, \gamma) \neq 0$$



but for

$$Y := S^3 \setminus \text{three unlinked circles}$$

$$\mu^y(\alpha, \beta, \gamma) = 0$$

03

AN APPLICATION TO ALGEBRAIC GEOMETRY



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The Donovan - Wemyss Conjecture

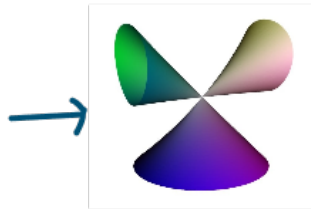
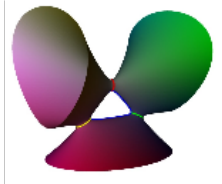
3D compound Du Val
singularities + cond.

Donovan - Wemyss

Finite-dim. algebras

$$p: X \rightarrow \text{Spec}(R)$$

$\uparrow$ crepant res. $\uparrow$ cDV sing



2 dim
analogue

Conj (Donovan - Wemyss 2013)

$$\Delta(R_1, p_1) \stackrel{\text{der}}{\sim} \Delta(R_2, p_2) \iff R_1 \cong R_2$$

$\Delta(R, p)$: contraction algebra

Wemyss 2018, Hwa - Keller 2024, August 2020

Algebraic
Problem



DW Conjecture

Solved in 2022 using A_∞-structures
in joint work with F. Muro

2024: New proof by Karaszyński - Lepri - Wemyss
also using A_∞-structures!